# Yang Runming

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Research interests: LLM post-training; Knowledge distillation; Efficient reasoning; Low-bit quantization

## EDUCATION

**The University of Hong Kong** Oct. 2025 - Present
Ph.D. student in Electrical and Electronic Engineering — Advisor: Prof. Ngai Wong; expected Sep. 2029

**Tsinghua University** Sep. 2022 - Jun. 2025
M.Eng. in Big Data Engineering — Advisor: Prof. Yujiu Yang; GPA: 3.73/4.00

**Huazhong University of Science and Technology** Sep. 2017 - Jun. 2021
B.Eng. in AI and Automation — Science and Engineering Cross-Innovation Experimental Class

## RESEARCH INTERNSHIP

**Tencent** Nov. 2023 - Jun. 2025
Research Intern

- Built data-synthesis pipelines for image-text content moderation and used the resulting data for model post-training and evaluation.
- Deployed models in tiers in production, fine-tuning large models on the full dataset and distilling smaller models using filtered data, improving throughput and reducing false positives over the previous solution.
- Researched large/small model collaboration and lightweight modeling, contributing to two academic papers and one company patent.

## ONGOING RESEARCH

**On-policy Distillation: Thinking-Mode Transfer and Subsequent RL** HKU 2026.09 - 2026.10

Qwen3 0.6B / 1.7B / 4B students and an 8B teacher, with comparisons of teacher prompting, generation budgets, and stopping behavior. Accuracy figures use MATH-500 avg@4.

- **Thinking-mode teachers overcome saturation in standard OPD gains.** **Gains from non-thinking teachers decline with student size**, reaching zero at 4B. With a thinking-mode teacher, 1.7B / 4B still gain 13.3 / 9.5 percentage points over their base models at a 16K budget.
- **Behavioral evidence supports transfer of the student's own thinking mode.** At 32K, 1.7B / 4B reach 90.9% / 95.2%, **approaching their own thinking-mode performance**, with similar results from self-distillation at all three sizes. Per-question accuracy profiles align more closely with the student's thinking mode and correlate significantly less with the external teacher, supporting gains from accessing existing reasoning capabilities.
- **Distinguished solving a problem from delivering an answer on time.** After truncation at 2K, allowing 32 additional tokens for a **forced answer** raises 1.7B accuracy from 17.8% to 72.8%, showing that OPD raises the non-thinking accuracy ceiling while also causing length inflation. Post-answer **p(EOS) and cross-continuation probes** show that the teacher still favors continuation after the student's answer, indicating that stopping behavior does not transfer alongside the thinking mode.
- **Characterized the benefits of OPD+RL and length control.** Under the same downstream RL setup, 1.7B starting from OPD reaches 87.3% at a 16K budget, versus 81.7% for direct RL. Short-run RL shows that rollout length affects the **trade-off between length compression and accuracy ceilings**. Propose reward adjustment using an adaptive length threshold at fixed rollout length to reduce redundant generation while preserving the reasoning needed for difficult problems.

## PUBLICATIONS

* Equal contribution.

[1] T. Wu*, **R. Yang***, T. Liu, J. Wang, N. Wong. *Revisiting Model Interpolation for Efficient Reasoning.* **ACL 2026 (Main)** [Paper] [Code]
Interpolates fast- and slow-thinking models without training to balance reasoning quality and chain-of-thought length.

[2] T. Liu*, T. Wu*, **R. Yang**, S. Sun, J. Wang, Y. Yang. *ProFit: Leveraging High-Value Signals in SFT via Probability-Guided Token Selection.* **ACL 2026 (Findings)** [Paper] [Code]
Uses current-policy information to guide token selection and relax token-level alignment during supervised fine-tuning.

[3] T. Wu*, Y. Cheng*, C. Ding*, **R. Yang**, X. Feng, W. Zhou, Z. Liu, N. Wong. *Can We Trust LLMs on Memristors? Diving into Reasoning Ability under Non-Ideality.* **COLM 2026** [Paper]
Simulates how memristor non-idealities affect LLM reasoning and evaluates training-free mitigation strategies.

[4] H. Xiao, Q. Yang, D. Xie, W. Xu, Z. Su, **R. Yang**, H. Liu, W. Zhou, Z. Liu, N. Wong. *Exploring Layer-wise Information Effectiveness for Post-Training Quantization in Small Language Models.* **ACL 2026 (Findings)** [Paper] [Project]
Uses layer-wise information effectiveness to allocate mixed precision across layers for low-bit quantization of small language models.

[5] T. Wu, C. Tao, J. Wang, **R. Yang**, Z. Zhao, N. Wong. *Rethinking Kullback-Leibler Divergence in Knowledge Distillation for Large Language Models.* **COLING 2025 (Oral)** [Paper] [Code]
Compares forward and reverse KL divergence for LLM distillation and introduces Adaptive KL.

[6] **R. Yang**, T. Wu, Y. Yang. *LoCa: Logit Calibration for Knowledge Distillation.* **ECAI 2024** [Paper]
Calibrates teacher logits for misclassified samples during knowledge distillation.

## PREPRINTS AND SUBMITTED MANUSCRIPTS

* Equal contribution.

[P1] **R. Yang***, T. Wu*, J. Li, P. Hu, Y.-C. Wu, N. Wong, Y. Yang. *Shadow-FT: Tuning Instruct Model via Training on Paired Base Model.* **Submitted to NAACL 2027** [arXiv] [Code]
Transfers fine-tuning updates from a base model to its paired instruction-tuned model.

[P2] H. Xiao*, **R. Yang***, Q. Yang*, W. Xu, Z. Li, Y. Su, Z. Liu, H. Yang, N. Wong. *PTQTP: Post-Training Quantization to Trit-Planes for Large Language Models.* **Submitted to NAACL 2027** [arXiv] [Project]
Decomposes LLM weights into two ternary trit-planes with lookup-table kernels for efficient inference.

[P3] T. Wu, **R. Yang**, T. Liu, J. Wang, Z. Xu, N. Wong. *Timber: Training-free Instruct Model Refining with Base Model via Effective Rank.* **Submitted to NAACL 2027** [arXiv] [Code]
Uses effective rank to analyze post-training weight changes and refine instruction-tuned models without additional training.

[P4] T. Liu*, T. Wu*, M. Zheng, X. Luo, **R. Yang**, X. Yang, J. Wang, Y. Yang. *Simple-OPD: Demystifying Warm-up for On-policy Distillation.* **Submitted to NAACL 2027** [arXiv] [Code]
Studies warm-up for on-policy distillation and initializes students with LoRA on teacher-generated chains of thought.

[P5] Q. Yang, **R. Yang**, H. Xiao, W. Xu, J. Chen, H. Liu, C. Ding, R. Hu, Y.-C. Wu, N. Wong. *FluxBin: Flexible LUT-based Ultra-low-bit LLM Inference by Algorithm-Kernel Synergy.* **Submitted to NAACL 2027** [arXiv] [Code]
Co-designs ultra-low-bit post-training quantization and lookup-table CUDA kernels for efficient LLM inference.

[P6] Y. Zhang, T. Wu, **R. Yang**, T. Liu, J. Wang, Y. Yang. *From Answers to Obligations: Benchmarking Financial Instruction Following with FinIF.* **Submitted to EMNLP 2027**
Studies instruction-following evaluation in financial tasks, focusing on whether model responses satisfy task requirements.

[P7] Z. Li*, Y. Su*, S. Wang, **R. Yang**, C. Xie, A. Liu, M. Li, J. Cao, Y. Xie, N. Wong, H. Yang. *Quantization Meets Reasoning: Exploring and Mitigating Degradation of Low-Bit LLMs in Mathematical Reasoning.* [arXiv]
Analyzes step-level reasoning errors in quantized LLMs and uses targeted preference pairs to improve 4-bit mathematical reasoning.

[P8] **R. Yang***, T. Wu*, J. Wang, P. Hu, Y.-C. Wu, N. Wong, Y. Yang. *LLM-NEO: Parameter Efficient Knowledge Distillation for Large Language Models.* [arXiv] [Code]
Combines low-rank adaptation and knowledge distillation for parameter-efficient knowledge transfer from large teachers to compact students.